

Operations Research & Applications programming

Ch. 1

Introduction To Operations Research

Theory

- ❑ **Definitions:** Operations Research (OR) is an interdisciplinary branch of applied mathematics and formal science that uses methods like *mathematical modeling*, statistics, and *algorithms* to arrive at optimal or *near optimal solutions* to complex problems.
- ❑ The formal activities of *Operations Research (OR)* were initiated in England during World War II when a team of British scientists set out to make decisions regarding *the best utilization of war material*.
- ❑ Today, **OR** is a dominant and indispensable **decision making tool**.

Introduction

- *This chapter*, will familiarize you with the basic terminology of *operations research*, including *mathematical modeling*, *feasible solutions*, *optimization*, and *iterative Computations*.
- The chapter also emphasizes that, while mathematical modeling is a cornerstone of **OR**, *intangible* (unquantifiable) *factors* (such as human behavior) must be accounted for in the final decision.

What is the model of OR

- *Example 1.1:* The *Bader* garment company manufactures men's shirts and women's blouses for *Mac discount stores*. *Mac* will accept all the production supplied by *Bader*. The production process includes cutting, sewing and packaging. *Bader* employs 25 workers in the cutting department, 35 in the sewing department and 5 in the packaging department.

What is the model of OR

- The factory works one 8-hour shift, 5 days a week. The following table gives the time requirements and the profits per unit for the two garments:

Minutes per unit

Garment	Cutting	Sewing	Packaging	Unit profit(\$)
Shirts	20	70	12	8.00
Blouses	60	60	4	12.00

- Determine the optimal weekly production schedule for Bader.

What is the model of OR

□ *Solution:* Assume that Baders produces x_1 shirts and x_2 blouses per week.

$$\text{Profit got} = 8 x_1 + 12 x_2$$

$$\text{Time spent on cutting} = 20 x_1 + 60 x_2 \text{ mts}$$

$$\text{Time spent on sewing} = 70 x_1 + 60 x_2 \text{ mts}$$

$$\text{Time spent on packaging} = 12 x_1 + 4 x_2 \text{ mts}$$

□ The *objective* is to find x_1, x_2 so as to *maximize* the profit

$$Z = 8 x_1 + 12 x_2 \text{ satisfying the } \textbf{constraints} :$$

What is the model of OR

$$20 x_1 + 60 x_2 \leq 25 \times 40 \times 60 = 60000$$

$$70 x_1 + 60 x_2 \leq 35 \times 40 \times 60 = 84000$$

$$12 x_1 + 4 x_2 \leq 5 \times 40 \times 60 = 12000$$

$$x_1, x_2 \geq 0, \text{ integers}$$

- ❑ This is a *typical optimization problem*.
- ❑ Any values of x_1, x_2 that satisfy all the constraints of the model is called a *feasible solution*.
- ❑ We are interested in finding the *optimum feasible solution* that gives the maximum profit while satisfying all the constraints.

What is the model of OR

- More generally, an **optimization** problem looks as follows:
- Determine the **decision variables** $x_1, x_2, \dots, x_n$ so as to **optimize** an **objective function**

$$f(x_1, x_2, \dots, x_n)$$

satisfying the **constraints** :

$$g_i(x_1, x_2, \dots, x_n) \leq b_i \quad (i = 1, 2, \dots, m).$$

Phases of OR study

- ❑ The Principle phases for implementation **OR** in practice include:
 - Definition of the problem.
 - Construction on the model.
 - Solution of the model.
 - Validation of the model.
 - Implementation of the solution.

Linear Programming Problem LPP “Ch2”

- ❑ An **optimization** problem is called a **Linear Programming Problem (LPP)** when the objective function and all the constraints are linear functions of the decision variables $x_1, x_2, \dots, x_n$.
- ❑ We also include the “non-negativity restrictions”, namely $x_j \geq 0$ for all $j = 1, 2, \dots, n$.
- ❑ Thus a typical **LPP** is of the form :

Linear Programming Problem LPP

Optimize (i.e. Maximize or Minimize)

$$Z = c_1 x_1 + c_2 x_2 + \dots + c_n x_n$$

subject to the constraints:

$$a_{11} x_1 + a_{12} x_2 + \dots + a_{1n} x_n \leq b_1$$

$$a_{21} x_1 + a_{22} x_2 + \dots + a_{2n} x_n \leq b_2$$

...

$$a_{m1} x_1 + a_{m2} x_2 + \dots + a_{mn} x_n \leq b_m$$

$$x_1, x_2, \dots, x_n \geq 0$$

Linear Programming Problem LPP

- ❑ A LPP satisfies the two properties :
Proportionality and additivity.
- ❑ Proportionality means the contributions of each decision variable in the objective function and its requirements in the constraints are directly proportional to the value of the variable.

Linear Programming Problem LPP

- **Additivity** stipulates that the total contributions of all the variables in the objective function and their requirements in the constraints are the direct sum of the individual contributions or requirements of each variable.

Linear Programming Problem LPP

- ❑ We shall first look at *formulation of some LPPs*,
- ❑ *Graphically solve* some LPPs involving two decision variables,
- ❑ *Study* some mathematical preliminaries regarding the solutions of LPPs,
- ❑ Finally look at the *Simplex method* of solving a LPP.

Tow - Variable LP Model

- ❑ ***Example 1.2*** Wild West produces two types of cowboy hats. Type I hat requires twice as much labor as a Type II. If all the available labor time is dedicated to Type II alone, the company can produce a total of 400 Type II hats a day. The respective market hats are 150 and 200 hats per day limits for the two types of. The profit is \$8 per Type I hat and \$5 per Type II hat.
- ❑ Formulate the problem as an LPP so as to maximize the profit.

Tow - Variable LP Model

- ***Solution:*** Assume that Wild West produces x_1 type I hats and x_2 type II hats per day.

$$\text{Per day Profit got} = 8 x_1 + 5 x_2$$

- Assume the time spent in producing one type II hat is c minutes.

Labor time spent is $(2 x_1 + x_2) c$ minutes

- The **objective** is to find x_1, x_2 so as to maximize the profit $z = 8 x_1 + 5 x_2$

Tow - Variable LP Model

satisfying the **constraints**:

$$(2x_1 + x_2) \leq 400$$

$$x_1 \leq 150$$

$$x_2 \leq 200$$

$$x_1, x_2 \geq 0, \text{ integers}$$

Tow - Variable LP Model

- *That is:* The **objective** is to find x_1, x_2 so as to maximize the profit $Z = 8 x_1 + 5 x_2$ satisfying the **constraints**:

$$(2 x_1 + x_2) \leq 400$$

$$x_1 \leq 150$$

$$x_2 \leq 200$$

$$x_1, x_2 \geq 0, \text{ integers}$$

Some Application of LP

Feed Mix problem

- ❑ The manager of a milk dairy decides that each cow should get at least 15, 20 and 24 units of nutrients A, B and C respectively. Two varieties of feed are available. In feed of variety 1(variety 2) the contents of the nutrients A, B and C are respectively 1(3), 2(2), 3(2) units per kg. The costs of varieties 1 and 2 are respectively 2 Rs. and 3 Rs. per kg.
- ❑ How much of feed of each variety should be purchased to feed a cow daily so that the *expenditure is least?*

Some Application of LP

Trim Loss problem

- ❑ A company has to manufacture the circular tops of cans. Two sizes, one of diameter 10 cm and the other of diameter 20 cm are required. They are to be cut from metal sheets of dimensions 20 cm by 50 cm. The requirement of smaller size is 20,000 and of larger size is 15,000.
- ❑ *The problem is* : how to cut the tops from the metal sheets so that the number of sheets used is a minimum. Formulate the problem as a LPP.
- ❑ *soln: $Z = x_1 + x_2 + x_3$, subject to $6x_1 + 2x_2 + 6x_3 \geq 20000$, $2x_2 + x_3 \geq 15000$, $x_1, x_2, x_3 \geq 0$ and integer, where x_1, x_2, x_3 be the number of sheets cut according to first, second and third pattern, respectively.*